Development of a Geometric Model of the Heart and Chest for Multichannel Electrical Impedance Computer Cardiography Technology

Alexey Tikhomirov
Bauman Moscow State Technical University
Moscow, Russia
tikhomirov.an@bmstu.ru

Andrey Briko
Bauman Moscow State Technical University
Moscow, Russia
briko@bmstu.ru

Nikolay Seleznev
Bauman Moscow State Technical University
Moscow, Russia
seleznev.nv@bk.ru

Sergey Shchukin
Bauman Moscow State Technical University
Moscow, Russia
0000-0002-9890-5267

Anton Levando
Bauman Moscow State Technical University
Moscow, Russia
anton-leva@yandex.ru

Mikhail Murashko
Bauman Moscow State Technical University
Federal Service for Surveillance in Healthcare
Moscow, Russia
0000-0002-4426-0088

Abstract—Impedance cardiography methods are inexpensive and convenient, they allow to get hemodynamic characteristics in real time. Electrical impedance methods for the heart reseach are met with distrust. This is due to the fact heart stroke volume is determined by empirical ratios, which do not always reflect the formation mechanisms of the electrical impedance signal in seriously ill patients [1]. The mathematical models use, both numerical and analytical, gave the development of new impedance cardiography (i.e. ICG) methods [2]. In this study, the influence of the heart and chest geometric model parameters on the results of calculations in precardial electroimpedance heart mapping was evaluated. In the work, a semi-infinite chest model was compared with real geometry obtained from MRI data. A comparison was also made of a real heart 3D model according to MRI and a sphere model approximating the heart blood. Studies have shown that the error in electrical impedance signal modeling brought in by using a semi-infinite chest model and simplification to the sphere is significant.

Keywords—Impedance cardiography, 3D model heart, approximation, electrical impedance computer cardiography.

I. Introduction

In the cardiovascular diseases (i.e. CVD) diagnosis and observation during treatment, an important role is played by the cardiac stroke volume (i.e. SV) evaluation. This can be done by various methods, such as CT, MRI, ultrasound. However, only the electrical impedance cardiography allows for non-invasive monitoring and frequent periodic evaluation of SV.

The classical ICG method developed by Kedrov, Kubicek, and Shramek used empirical equation under certain conditions and gave a correlation coefficient up to 0.81 (Fullen, 1982) [3], but became less reliable in the case of in patients with severe cardiorespiratory system disorders [1].

Today ICG methods are used in clinical practice in devices such as Niccomo and Cardioscreen from Medis, several devices manufactured by Physioflow and others.

The reported study was funded by RFBR according to the research project No 18-29-02042.

Using empirical ratios in the calculation is a major drawback ICG methods. Numerous studies on the determination of stroke volume hearts in various categories of patients showed insufficient accuracy for clinical practice [4], which, in particular, formed a negative attitude towards rheocardiography in the medical community.

The technology of electrical impedance tomography is actively developing, as it is relatively inexpensive, safe and allows long term monitoring of the physiological function [5]. A major EIT study focused on pulmonary investigation [5, 6]. The spatial resolution provided by EIT currently does not allow direct estimation of both the volumetric characteristics of the activity of the heart and the movement of the walls of the ventricles [7, 8, 9, 10, 11, 12].

The new technologies development, including mathematical and numerical modeling use, has given a new round to the ICG methods development. In modern studies, the empirical expressions of the SV definition according to the impedance change data are replaced by the dependences obtained during analytical modeling [13] and numerical simulation of the impedansometry task.

Using a priori information from MRI and CT data to determine the parameters of mathematical models can improve the accuracy of determining the movements of the walls of the ventricles. In this case, firstly, it is important to understand the mechanisms of formation of electric impedance precordial signals, and secondly, the choice of a mathematical model. The mathematical model should correspond to the mechanisms of the formation of the electrical impedance signal, but not be overly complicated [13,14].

The geometric models used in the simulation are different. In the works [2, 6] cylinder as a model of the chest is used. Sokomoto (1979) [7], Ulbrich (2014) [17] used simple geometric shapes to define the chest and heart, some authors used real geometry according to MRI / CT data. For ICG precordial methods, semi-infinite space models were used. The heart itself in the model was usually represented in the form of real geometry or a sphere approximating the volume of heart blood.

Authorized licensed use limited to: Florida Institute of Technology. Downloaded on May 12,2025 at 00:38:09 UTC from IEEE Xplore. Restrictions apply.

The goal of this study was to estimate the error in precordial measurements caused by the use of the simplest geometric chest and heart forms in the simulation of electrical impedance measurements [18]. Comparable models of the chest, represented by a semi-infinite half-space and real chest geometry, were obtained from an MRI study. We also compared a heart blood 3D model obtained by MRI data and a sphere obtained by approximating a 3D model using the least squares method.

II. Materials and Methods

A. Geometric Model

A geometric 3D chest and heart model was obtained according to an MRI study performed at the Bakulev's Research Institute on the Siemens Magnetom Avanto.

A 3D model of the chest was obtained by extruding the profile of soft tissues and the fat layer (red and green lines in Fig. 1). Then, a heart model was placed in the obtained chest model. Segmentation of blood in the heart on MRI images was performed in 3DSlicer, then exported to a stl file (Fig. 2). Additional processing to account for mesh defects was performed in the MeshLab program.

The simulation was carried out with the values of the electrical resistivity of the tissues at a frequency of 100 kHz, which corresponds to the probe frequency of the REO-32 [14] apparatus used by us for experimental studies. Fat tissue was isolated in a separate layer with a specific electrical resistance of 20 Ohm * m (Table I). Myocardial resistivity 4.6 Ohm * m. Muscle tissue has a high anisotropy of electrical resistivity and varies greatly according to [19] in the range from 1.5 to 25 Ohm * m, according to 2.7 Ohm * m, but for the chest it is recommended [19] to use a value 4.63 Ohm * m. At this stage of the study, electrical impedance measurements are performed on exhalation or cardiocycles are released in the exhalation phase. The value of the specific resistance of the lungs on the exhale is 3.68 Ohm * m. Since the values of the specific resistance of the myocardium, muscle tissue and lungs during the exhalation phase are comparable, therefore, in the geometric model they were combined into a single volume with the specific resistance of 4.2. The recommended value of 4.63 Ohm * m was corrected to 4.2 Ohm * m based on measurements from the thoracic channel positioned on the surface of the chest to the left of the heart [14].

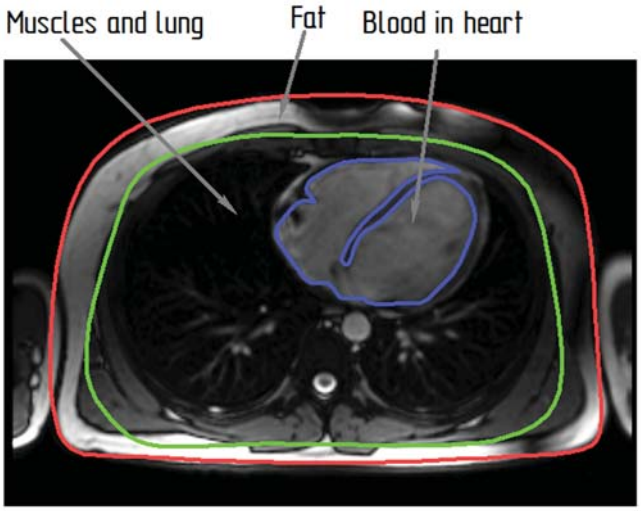

Fig. 1. A geometric model of the chest (red) and blood (blue) in the heart according to MRI

The heart blood approximation by an equivalent sphere was carried out by the method of least squares of the distances from the sphere to the points of the blood volume reconstructed surface in accordance with the criterion:

$$\delta = \sqrt{\sum_{i=0}^{n}\left(\sqrt{(x_0-x_i)^2+(y_0-y_i)^2+(z_0-z_i)^2}-R_{sph}\right)}$$

where (x_0, y_0, z_0) — the coordinates of the surface sphere center; (x_i, y_i, z_i) — coordinates of the i-th point of the 3D heart model; R_{sph} — the sphere radius.

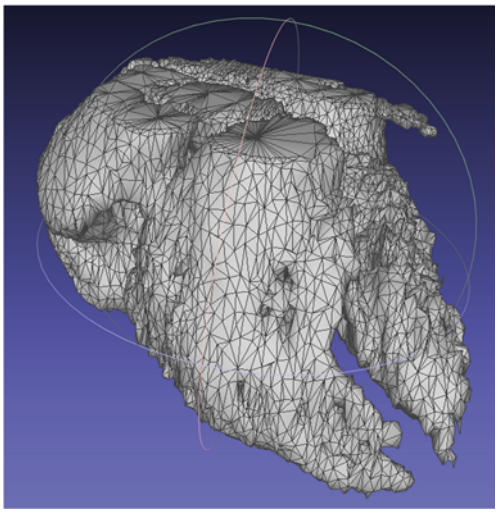

Fig. 2. 3D model of blood in the heart according to MRI

B. Modeling

The modeling was carried out in the finite element analysis environment ANSYS. The parameters of the tissue resistivity values were in accordance with Table I.

TABLE I. Resistivity values at 100 kHz used in the model study

Tissue	Resistivity values, Omh*m	Resistivity value used in the simulation, Omh*m
Fat tissue	20 [20]	20
	20 - 50 [1]	
	25 [19]	
Blood (Hct=50)	1.35 [21]	1.35
Myocardium	4.6 [20]	4.2
Muscles	2.7 [20]	
	1.5 – 25 [19]	
Lung (deflated)	3.68 [20]	
Lung	1.6 – 10 [1]	
Human thorax (average)	4.63 [19]	

The electrode system was located in the 5th channel region using precordial radial electrical impedance mapping (Fig. 3) since in this area, the difference between the real chest geometry with the compared semi-infinite model is expressed. The values of the basic impedance of Z_{base} were evaluated. A resistivity change of the tissues surrounding the heart caused by pulse blood filling of the soft tissues was not made.

Authorized licensed use limited to: Florida Institute of Technology. Downloaded on May 12,2025 at 00:38:09 UTC from IEEE Xplore. Restrictions apply.

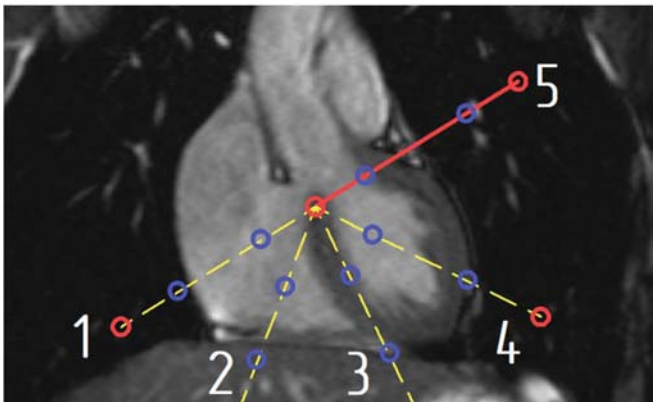

Fig. 3. The arrangement of the electrode system of the 5th channel by the method of precardial impedance computer cardiography

The distribution of current density in the simulated region is not uniform and the current is concentrated in the region of the electrode system. To increase the accuracy of the simulation, a hemisphere-shaped region was selected near the electrode system. In Fig. 4, near the electrodes (red – current electrodes, blue – measurement electrodes), the region (highlighted in yellow), including part of the adipose tissue, soft tissues and heart blood (red dotted line) has a characteristic mesh size reduced compared to the rest of the model. The total number of mesh elements ranged from 2 million to 3.5 million depending on the model.

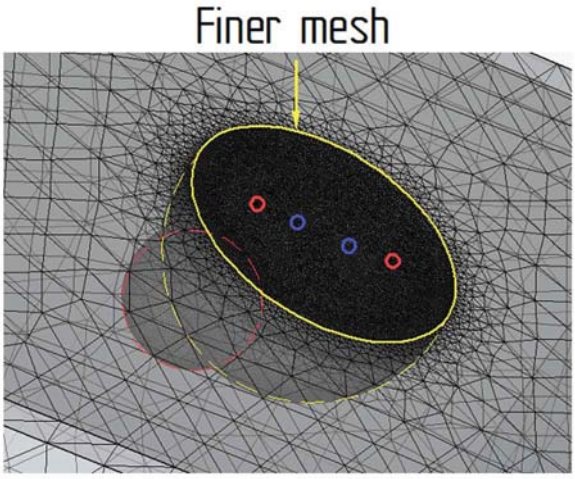

Fig. 4. The area around the electrode system with improved mesh

III. RESULTS

To separate the error caused by the difference in the chest geometry from the semi-infinite half-space and the error caused by the difference in the real 3D geometry of the heart blood from the approximating sphere, the following models were considered.

First, the realistic model consisting of a realistic chest model and the 3D heart model (Fig. 5a).

Secondly, the realistic chest model and the equivalent sphere placed inside, approximating the heart blood (Fig. 5c).

Thirdly, the semi-infinite environment with the 3D heart model (Fig. 5b).

These methods were compared with the currently used [14] model of a sphere in a semi-infinite half-space (Fig. 5d).

The results of modelling the base impedance Z_{base} are shown in Table II.

The basic impedance value obtained from a realistic 3D model can be considered an experimental result, and with a fixed Z_{base} equal to Z_{base_a}, the equivalent electrical resistivity of the soft tissues was determined for models b, c, d (Fig. 5).

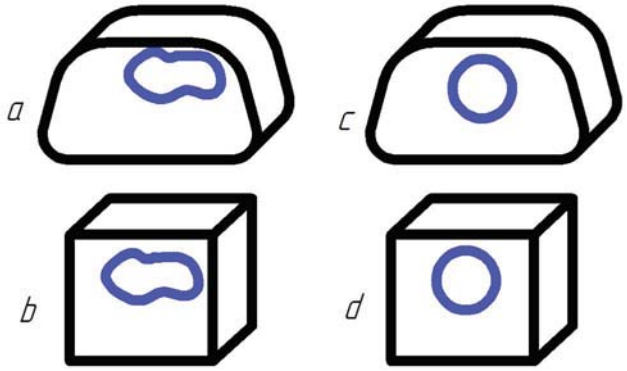

Fig. 5. Different models of the chest and blood in the heart: a – realistic chest model with 3D heart model, b – half-space with 3D heart model, c - realistic chest model with equivalent sphere, d - half-space with equivalent sphere

The dependences of the base impedance value on the resistivity of soft tissues were also obtained during the simulation (Fig. 6). From these dependencies, the equivalent specific resistance of soft tissues was determined (Table III).

TABLE II. VALUES OF BASIC RESISTANCE FOR DIFFERENT MODELS OF THE CHEST (REALISTIC AND HALF-SPACE) AND BLOOD IN THE HEART (3D MODEL AND EQUIVALENT SPHERE)

Z_{base}, Ohm		
	Z_{base_a} = 99.45	Z_{base_c} = 100.18
	Z_{base_b} = 97.47	Z_{base_d} = 98.06

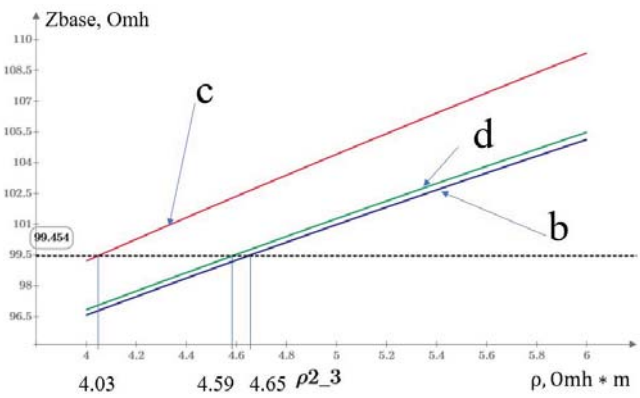

Fig. 6. Dependences of the value of the base impedance on the resistivity of soft tissues for models b,c,d

Authorized licensed use limited to: Florida Institute of Technology. Downloaded on May 12,2025 at 00:38:09 UTC from IEEE Xplore. Restrictions apply.

TABLE III. VALUES OF RESISTIVITY FOR DIFFERENT MODELS OF THE CHEST (REALISTIC AND HALF-SPACE) AND BLOOD IN THE HEART (3D MODEL AND EQUIVALENT SPHERE)

ρ_{eq} ,Ohm*m		
	$\rho_{eq} = 4.2$	$\rho_{eq} = 4.03$
	$\rho_{eq} = 4.65$	$\rho_{eq} = 4.59$

IV. DISCUSSION

The deviation in basic impedance for models 2-4 from the value of the realistic model basic impedance was reaches up to 1.5%

Thus, in accordance with Table III the error in determining the equivalent resistivity is from -5% to 9%. Moreover, according to [22] with such an error in determining the resistivity of soft tissues, the error in determining the movement of the heart ventricles wall in the electrode system region can reach 12 %.

V. CONCLUSION

The research showed that when simulating electrical impedance measurements, using a homogeneous half-space as a geometric chest model and approximating the heart blood with an equivalent sphere can lead to an error in determining the linear displacements of the heart ventricles wall in the electrode system region up to 12 %. The use of a spherical model in [13,14] can lead to an additional error in determining the linear displacements of the walls of the heart.

Thus, it is necessary to use geometric models of the heart that are more consistent with real geometry, such as an ellipsoid or ovoid.

REFERENCES

[1] S. Grimnes, O. G. Martinsen, Bioimpedance and bioelectricity basics. CA: San Diego, 2008.

[2] A. de Sitter, R. M. Verdaasdonk and T. J. C. Faes, "Do mathematical model studies settle the controversy on the origin of cardiac synchronous trans-thoracic electrical impedance variations? A systematic review", Physiological Measurement vol. 37, pp. 88-108, August 2016.

[3] H. D. Fuller, Validity of cardiac output measurement by thoracic impedance: A meta-analysis. Clin. Invest. Med. vol. 15, pp. 103–112, 1992

[4] P. J. Riu, "Electrical bioimpedance methods: application to the medicine and biotechnology", NewYork Academy of Science, 1999.

[5] B. H. Brown, "Electrical impedance tomography (EIT): A review," J. Med. Eng. Technol., vol. 27, no. 3, pp. 97–108, 2003.

[6] H. Luepschen, D. Van Riesen, L. Beckmann, K. Hameyer, and S. Leonhardt, "Modeling of fluid shifts in the human thorax for electrical impedance tomography," IEEE Trans. Magn., vol. 44, no. 6, pp. 1450–1453, 2008.

[7] T. Kriz and Z. Roubal, "Practical application of electrical impedance tomography and electrical resistive tomography," in 2016 Progress In Electromagnetics Research Symposium, PIERS 2016 - Proceedings, 2016, pp. 1499–1503.

[8] S. Babaeizadeh, D. H. Brooks, and D. Isaacson, "3-D electrical impedance tomography for piecewise constant domains with known internal boundaries," IEEE Trans. Biomed. Eng., vol. 54, no. 1, pp. 2–10, Jan. 2007.

[9] M. Kircher, R. Hattiangdi, R. Menges, and O. Dossel, "Influence of background lung tissue conductivity on the cardiosynchronous EIT signal components: A sensitivity study," Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS, vol. m, pp. 1547–1550, 2019.

[10] T. Rymarczyk and K. Kania, "Inverse Problem Solution for Model with Lungs and Heart in EIT," 2019 Appl. Electromagn. Mod. Eng. Med., pp. 180–183, 2019.

[11] Y. Zhou, X. Dong, F. Fu, and X. Shi, "Fast High-Precision Electrical Impedance Tomography System for Real-Time Perfusion Imaging," IEEE Access, vol. 7, pp. 61570–61580, 2019.

[12] Y. Wu et al., "A High Frame Rate Wearable EIT System Using Active Electrode ASICs for Lung Respiration and Heart Rate Monitoring," vol. 65, no. 11, pp. 3810–3820, 2018.

[13] D. P. Timokhin, "Development of a Biotechnical System for Multichannel Electrical Impedance Mapping of the Biomechanical Activity of the Heart", Dissertation for Master's Degree in Technical Sciences [in Russian], Bauman Moscow State Technical University (2014).

[14] A. N. Tikhomirov, S. I. Shchukin, S. Leonhardt, A. K. Volkov and M. A. Murashko, "Multichannel Electrical Impedance Methods for Monitoring Cardiac Activity Indicators", Biomedical Engineering, vol. 52, no. 6, pp. 365-370, March 2019.

[15] M. X. Kuang, Q. J. Xiao, C. Y. Cui, N. Z. Kuang, W. Q. Hong and A. R. Hu "Mechanism of the formation for thoracic impedance change" Annals of Biomedical Engineering, vol. 38, issue 3, pp. 1007–1016

[16] K. Sakamoto, K. Muto, H. Kanai and M. Lizuka "Problems of impedance cardiography" Medical & Biological Engineering & Computing, vol. 17, pp. 697–709.

[17] M. Ulbrich, J. Muhlsteff, S. Leonhardt and M. "Walter Influence of physiological sources on the impedance cardiogram analyzed using 4D FEM simulations", Physiological Measurement, vol. 35, pp. 1451–1468, June 2014.

[18] F. Yang, R.A. Patterson, "Simulation Study on the Effect of Thoracic Conductivity Inhomogeneities on Sensitivity Distributions", Ann Biomed Eng, vol. 36, pp. 762–768, 2008.

[19] S. Rush, J.A. Abildskov, "Resistivity of body tissues at low frequencies", Circulation Research, vol. 12, pp. 40–50, 1963.

[20] P. A. Hasgal, F. Di Gennaro, C. Baumgartner, E. Neufeld, B. Lloyd, M. C. Gosselin, D. Payne, A. Klingenbock, N. Kuster, "IT'IS Database for thermal and electromagnetic parameters of biological tissue", Ver. 4.0, May 2018

[21] D. W. Hill, F. D. Thompson, "The effect of haematocrit on the resistiviry of human blood at 37 degrees C and 100 kHz", Medical and biological engineering, vol. 13, pp. 182-6, 1975.

[22] A. M. Tikhomirov, A. I. Malakhov, S. I. Shchukin, I. K Sergeev and A. K. Volkov, "Finite element analysis as a method to verify two layer mathematical model with spherical immersion", Biomedicine Radioengineering, vol. 7, pp. 9–12, 2015.

Authorized licensed use limited to: Florida Institute of Technology. Downloaded on May 12,2025 at 00:38:09 UTC from IEEE Xplore. Restrictions apply.